MTH219 2023S1 Exam

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* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

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SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH219

Examination - January Semester 2023

Fundamentals of Statistics and Probability

Thursday, 18 May 2023 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **SEVEN (7)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (Total 100 marks)

Question 1a

Strawberries are packed in boxes of 20. It is known that, on average, the probability that any particular strawberry being spoiled is p. Let S be the random variable representing the number of spoiled strawberries in a box.

Question 1a(i)

State, in the context of the question, two assumptions needed to model S by a binomial distribution. Analyze and discuss if you think that the assumptions were met.

(5 marks)

Question 1a(ii)

Assume that the two assumptions in Question 1a(i) do in fact hold. A box is selected at random and the probability that at most 1 strawberry (i.e. either 0 or 1 strawberry) is spoiled is 0.73584.

Show that $(1-p)^{19}(1+19p) = 0.73584$, where p is the probability that any particular strawberry is spoiled.

(4 marks)

Question 1a(iii)

Verify that $p = 0.05$ will meet the requirement of the equation in Question 1a(i).

(2 marks)

Question 1a(iv)

There are 100 boxes of strawberries in a delivery. A box will be returned if there are at least 4 spoiled strawberries. By applying the concepts from the binomial distribution, find the expected number of boxes which are returned.

(3 marks)

Question 1b

After a full charge, the time a particular brand of phone can last until the battery goes flat is exponentially distributed with average of 12 hours. The owner of the phone needs to go out for 6 hours.

Question 1b(i)

Apply a suitable statistical model to find the probability that the owner's phone will NOT run out battery during the time where he is out.

(4 marks)

Question 1b(ii)

Compute the probability that the owner's phone will run for another 6 hours, given that the phone has already lasted for 6 hours.

(5 marks)

Question 1b(iii)

Comment on the reasonableness of using exponential distribution as a model for T, the amount of time for a fully charged phone battery to run out of power.

(2 marks)

Question 2a

In a supermarket, the masses in grams of a carton of ice cream (X) have the distribution $X \sim N(500, 10^2)$ and the masses in grams of a banana (Y) have the distribution $Y \sim N(110, 25^2)$. X and Y are independent.

Define any random variables used in this question clearly.

Question 2a(i)

Determine the Upper Quartile, Q_3, Median, Q_2, Lower Quartile, Q_1 and the inter-quartile range of the mass of a banana, Y.

(5 marks)

Question 2a(ii)

Alex bought 10 bananas and 2 cartons of ice cream. The bananas were packed into a bag, and the ice-cream in another. Compute the probability that the ice-cream bag is more massive than the bag of bananas.

(5 marks)

Question 2a(iii)

A certain dessert recipe requires 0.5 carton of ice-cream and 5 bananas. Assume that the ice-cream carton and bananas are chosen randomly. Compute the probability that the total mass of 0.5 carton of ice-cream and 5 bananas falls between 700 and 850 grams.

(5 marks)

Question 2b

Hurricanes form regularly in the sea, but not many actually cause damage to civilization as only a few will actually make landfall each year. In a particular country, they experience an average of 3 hurricanes each year. Assume that the Poisson Distribution is a suitable statistical model for the Total number of Hurricanes in t years.

Question 2b(i)

State 2 assumptions of the homogeneous Poisson Distribution.

(2 marks)

Question 2b(ii)

Calculate the probability of more than 1 hurricane occurring in 2 years.

(4 marks)

Question 2b(iii)

On average the Country will take one month to recover from the damage of one hurricane.

The country was just hit by a hurricane. Apply a suitable probability model to find the probability that they will be ravaged by another hurricane while recovering from the current one.

Assume that one month is exactly 1/12 of a year.

(4 marks)

Question 3a

A group of 300 adults were asked on their preferences for cardiovascular exercises.

Here are a few facts gathered:

- 100 of them like Swimming.
- Half of them like Swimming OR Running (or both).
- 30 of them like Swimming AND Cycling.
- None of them like both Running and Cycling. (And by extension, no one likes all three)
- Liking Swimming is independent from liking Running

Let the events where they like Swimming, Running and Cycling be denoted by S, R and C respectively.

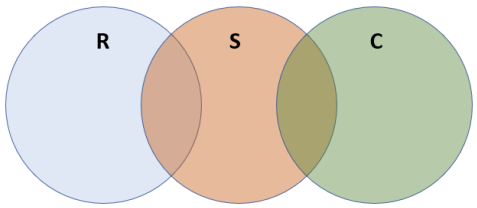


Figure Q3(a): Venn Diagram presenting the relationship between R, S and C

Question 3a(i)

Explain in words, why Regions R and C in Venn Diagram in Figure Q3(a) do not overlap.

(3 marks)

Question 3a(ii)

With the help of the Venn Diagram, find $P(S' \cap R')$.

Hint: $S' \cap R'$ means "Neither Swim nor Run".

(4 marks)

Question 3a(iii)

If $P(R) = \frac{1}{4}$, show that R and S are independent. Hint: Use Additive rule.

(6 marks)

Question 3a(iv)

Assume that all the surveyed adults like at least one of the three types of cardiovascular exercise (i.e. $n(R \cup S \cup C) = 300$). Find the value of $P(C)$ and explain your answer.

(6 marks)

Question 3b

Let X be a continuous random variable. It is given that $4P(X > k) = 5P(X \leq k)$ for a fixed constant k.

Question 3b(i)

Find $P(X > k)$.

Hint: $P(X > k) + P(X \leq k) = 1$.

(3 marks)

Question 3b(ii)

12 independent observations of X are random generated. Find the probability that there are more observations of X with values greater than k, than observation of X with values less than (or equal) k.

(3 marks)

Question 4a

When you are looking through social media one day, you found a post that claims that taking Fish Oil (Omega-3) 2-hours before sleeping will help improve the quality of sleep among young adults aged 20 to 30.

You want to investigate if this is true by doing an experiment with your classmates.

Question 4a(i)

<u>In the context of this question</u>, explain what/who is

(i) The Population;
(ii) The Sample.

Why do we collect samples and what are some of the considerations when doing so?

(8 marks)

Question 4a(ii)

Your classmates were asked to report their number of uninterrupted sleeping duration (in hours) after taking the fish oil.

The number of uninterrupted hours of sleep for 8 of your classmates are on Monday night: 5, 5, 6, 6, 7, 8, 10, 12.

Compute the mean and variance.

(6 marks)

Question 4b

John always receives two \$2 notes, three \$5 notes and two \$10 note from his parents as his weekly allowance. He decides to randomly select two of these notes to donate to charity this week. Let the total value of the two notes to be donated to charity be $\$X$.

Question 4b(i)

By considering all possible combinations of two notes, determine PMF of X.

Combinations	X	$P(X = x)$
2,2	4	
2,5 or 5,2	7	
5,5	10	
2,10 or 10,2	12	
5,10 or 10,5	15	
10,10	20	

<u>Table Q4(b)(i)</u>

(6 marks)

Question 4b(ii)

Calculate $E(X)$ and $Var(X)$.

(5 marks)

----- END OF PAPER -----

1(a) i. The probability that a strawberry is spoilt is independent. The total number of trials is fixed at 20.

ii. $S \sim Bin(20, p)$

$P(S \leq 1) = 0.73584$

$P(S=0) + P(S=1) = 0.73584$

$\binom{20}{0} p^0 (1-p)^{20} + \binom{20}{1} p (1-p)^{19} = 0.73584$

$(1-p)^{20} + 20p(1-p)^{19} = 0.73584$

$[(1-p) + 20p](1-p)^{19} = 0.73584$

$\therefore (1+19p)(1-p)^{19} = 0.73584$

iii. let $p = 0.05$

$1+19p = 1.95$

$1-p = 0.95$

$(1+19p)(1-p)^{19} = 1.95 \times 0.95^{19}$

$= 0.7358395$

$\simeq 0.73584$

iv. let X be the number of returned boxes

$X \sim Bin(100, P(S \geq 4))$

$P(S \geq 4) = 1 - P(S \leq 3)$

$= 1 - \text{pbinom}(3, 20, 0.05)$

$= 0.01590153$

expected number of returned boxes $= E(X)$

$= 100 \times P(S \geq 4)$

$= 1.59$ boxes

1(b) i. let T be the time the phone battery lasts

$$\mu = 12h$$
$$\Rightarrow \lambda = \frac{1}{12} h^{-1}$$
$$T \sim Exp(\tfrac{1}{12})$$

P(phone battery lasts while owner is out)

$$= P(T \geq 6h)$$
$$= 1 - P(T \leq 6h)$$
$$= 1 - pexp(6, \tfrac{1}{12})$$
$$= 0.6065307$$

ii. P(phone lasts another 6h given the phone already lasted for 6h)

$$= \frac{P(\text{phone lasts } 12h)}{P(\text{phone already lasts } 6h)}$$
$$= \frac{P(T \geq 12h)}{P(T \geq 6h)}$$
$$= \frac{1 - P(T \leq 12h)}{1 - P(T \leq 6h)}$$
$$= \frac{1 - pexp(12, \frac{1}{12})}{1 - pexp(6, \frac{1}{12})}$$
$$= 0.6065307$$

iii. As time passes by, the expected time for the phone battery to lose power actual reduces instead of remaining constant. Also, the memoryless property of the exponential distribution model cannot apply to battery life. Hence, the exponential model is unsuitable

2(c) i. $Y \sim N(110, 25^2)$

$Q_1 = 25^{th}$ percentile

$= \text{qnorm}(0.25, 110, 25)$

$= 93.13776$

$Q_2 = 50^{th}$ percentile

$= \text{qnorm}(0.50, 110, 25)$

$= 110$

$Q_3 = 75^{th}$ percentile

$= \text{qnorm}(0.75, 110, 25)$

$= 126.86224$

$IQR = Q_3 - Q_1$

$= 33.72448$

ii. $X \sim N(500, 10^2)$

let $Z = 2X - 10Y$

$E(Z) = 2E(X) - 10E(Y)$

$= 2(500) - 10(110)$

$= -100$

$Var(Z) = Var(2X - 10Y)$

$= 4Var(X) + 100\,Var(Y)$

$= 4 \times 10^2 + 100 \times 25^2$

$= 62900$

P(ice cream bag heavier than banana bag)

$= P(Z > 0)$

$= 1 - P(Z \leq 0)$

$= 1 - \text{pnorm}(0, -100, \sqrt{62900})$

$= 0.3450475$

2(a) iii. let $M = 0.5X + 5Y$

$E(M) = E(0.5X + 5Y)$

$= 0.5E(X) + 5E(Y)$

$= 800$

$Var(M) = Var(0.5X + 5Y)$

$= 0.25\,Var(X) + 25\,Var(Y)$

$= 0.25(100) + 25^3$

$= 15650$

$M \sim N(800, 15650)$

$P(700 \le M \le 850)$

$= pnorm(850, 800, \sqrt{15650}) - pnorm(700, 800, \sqrt{15650})$

$= 0.4432634$

2(b) i. Each event occurs 1 at a time.

The average number of events per unit time interval remains constant

ii. let X be the number of hurricanes in 2 years

$X \sim Po(6)$

$P(X > 1) = 1 - P(X \le 1)$

$= 1 - ppois(1, 6)$

$= 0.9826487$

iii. let Y be the number of hurricanes in 1 year

$Y \sim Po(3)$

let T be the time between 2 hurricane

$T \sim Exp(3)$

$P(\text{county rocked from another hurricane during recovery})$

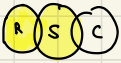
$= P(T \le \frac{1}{12})$

$= pexp(\frac{1}{12}, 3)$

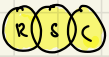
$= 0.2211992$

3(a) i. No one likes both running and swimming. Hence the regions R and C don't overlap.

ii. $\bar{S} \cap \bar{R}$ =

$R \cup S$ = 

$n(R \cup S) = 150$

$R \cup S \cup C$ = 

$n(R \cup S \cup C) = 300$

$$n(\bar{S} \cap \bar{C}) = n(R \cup S \cup C) - n(R \cup S)$$
$$= 300 - 150$$
$$= 150$$
$$P(\bar{S} \cap \bar{C}) = \frac{150}{300} = 0.5$$

iii. $P(R) = \frac{1}{4} \Rightarrow n(R) = P(R) \times 300 = 75$

$n(S) = 100 \Rightarrow P(S) = \frac{100}{300} = \frac{1}{3}$

$n(R \cup S) = n(R) + n(S) - n(R \cap S)$

$150 = 75 + 100 - n(R \cap S)$

$n(R \cap S) = 25$

$P(R \cap S) = \frac{25}{300} = \frac{1}{12}$

$P(R) . P(S) = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12}$

Since $P(R) . P(S) = P(R \cap S)$, then R and S are independent

iv.

$n(S \cap C) = 30$

$n(\bar{S} \cap \bar{R}) = 150$

$n(C) = n(S \cap C) + n(\bar{S} \cap \bar{R}) = 180$

$\therefore P(C) = \frac{180}{300} = \frac{3}{5}$

3b. i.

$$4P(X>k) = 5P(X\le k)$$

$$P(X>k) + P(X\le k) = 1$$

$$\Rightarrow P(X>k) = 1 - P(X\le k)$$

$$4(1 - P(X\le k)) = 5P(X\le k)$$

$$4 - 4P(X\le k) = 5P(X\le k)$$

$$9P(X\le k) = 4$$

$$P(X\le k) = \frac{4}{9}$$

$$P(X>k) = 1 - P(X\le k) = \frac{5}{9}$$

ii. let n be no. of observations with $X > k$

$$n \sim Bin(12, \tfrac{5}{9})$$

suppose $n > 12 - n$

$$2n > 12$$

$$n > 6$$

required probability $= P(n>6)$

$$= 1 - P(n\le 6)$$

$$= 1 - \text{pbinom}(6, 12, \tfrac{5}{9})$$

$$= 0.54265$$

4(a) i. Population refers the population of young adults aged 20 to 30.

Sample refers to the actual group of young adults whose quality of sleep is surveyed before and after the consumption of fish oil

We collect samples to perform hypothesis testing. The main consideration is to prevent biasness in the sampling. The main techniques are random sampling, systematic sampling.

ii. Mean $= \bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i = 7.375$

$var(X) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2 = 6.267857 \simeq 6.268$

4(b) i. There are 2x \$2, 3x\$5, 2x \$10

total no. of ways $= 7\times 6 = 42$

no. of ways to pick (2,2) $= 2\times 1 = 2$ $\Rightarrow P(X=4) = \frac{2}{42} = \frac{1}{21}$

no. of ways to pick (2,5), (5,2) $= 2\times 3 + 3\times 2 = 12$ $\Rightarrow P(X=7) = \frac{12}{42} = \frac{2}{7}$

no. of ways to pick (5,5) $= 3\times 2 = 6$ $\Rightarrow P(X=10) = \frac{6}{42} = \frac{1}{7}$

no. of ways to pick (2,10), (10,2) $= 2\times 2 + 2\times 2 = 8$ $\Rightarrow P(X=12) = \frac{8}{42} = \frac{4}{21}$

no. of ways to pick (5,10), (10,5) $= 3\times 2 + 2\times 3 = 12$ $\Rightarrow P(X=15) = \frac{12}{42} = \frac{2}{7}$

no. of ways to pick (10,10) $= 2\times 1 = 2$ $\Rightarrow P(X=20) = \frac{2}{42} = \frac{1}{21}$

ii $E(X) = \sum_{i=1}^{6} x_i P(X=x_i) = 11.14286$

$var(X) = \sum_{i=1}^{6}(x_i-\bar{x})^2 P(X=x_i) = 15.64626$